

INFORMATION TECHNOLOGY

Paper 9626/11 Theory

Key messages

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Candidates are reminded that answers need to be legible. Whilst the number of illegible answers remains very small, and every effort is made to read them, where answers cannot be read, examiners are unable to give marks.

Candidates are also reminded that this examination is intended to assess their knowledge of Information Technology to a high level of understanding. Therefore, it is a reasonable expectation that candidates will be able to use technical terms from across the syllabus accurately and employ suitably technical terms in their answers. In a small number of cases, candidates use slang terms as part of their answer. Whilst those in common use, such as 24/7, are acceptable, despite being outside the realms of the syllabus, candidates should avoid using slang terms to answer technical questions.

General comments

Previous versions of this report have commented on the improvement in how candidates are responding to key words. This improvement continued with this iteration of the examination. Candidates are now attempting to explain when asked to do so, rather than describe, as they would previously. However, candidates are reminded that the key word 'justify' means to give arguments in support of an action or choice. Many candidates respond to such questions by arguing both for and against.

Candidates would also benefit from further attention being paid to context. Context may be seen as giving a framework within which answers may be given. As such, context also provides a restriction, as certain answers may be acceptable for one scenario, but not for another.

Finally, candidates are reminded that where a specific number of answers are requested in the question, only that number of responses may be considered. If, for example, the question asks for two reasons why an action may be taken, only the first two answers given by the candidate may be awarded marks. Typically, where candidates join two concepts together using 'and', the concept after the word 'and' is treated as a second answer, unless the piece after the 'and' is clearly an explanation or clarification of the piece before the 'and'.

Comments on specific questions

Question 1

Variations on this question have appeared regularly over recent years and have rarely caused any issues for candidates. This cohort of candidates answered both parts well.

Question 2

- (a) Candidates were asked to describe the benefits. Because the word 'benefits' was in plural, candidates had to give more than one benefit. As a description is generally a one-mark answer, candidates had to give three separate descriptions for full marks to be awarded.

Most candidates scored at least two marks for this question. Typically, candidates focussed on the data being both relevant and up to date. Wrong answers were typically that the data would not be biased. This was not acceptable.

- (b) Candidates did not answer this question well. Many suggested that collecting data is a time-consuming business. This was not accepted, as it is not necessarily true. Candidates were not told which data the nurse was collecting, and so were unable to state with any certainty that data collection would take a long time.

Question 3

- (a) The first question asked candidates to describe how vishing obtains personal details. Due to the relative similarities between methods of collecting personal data, candidates were required to show clear understanding that the method involved a phone call from someone claiming to be an employee of the bank or asking for bank details. Without such a requirement, candidates could have listed a number of unspecific points for full marks, without ever showing any unique knowledge of vishing. This use of an initial point as a gateway marking point is likely to be repeated in the future.

Candidates who achieved the first mark typically then went on to achieve full marks for the question.

- (b) This part was intended to be a low demand question, and most candidates who suggested techniques achieved good marks. However, there were some misconceptions. Most notable of these was that the intended target should contact the bank to check on the veracity of the call. This is not a suitable method, as a vishing attack may take over a phone, so that any call goes to the attacker's phone. For the mark to be awarded, candidates had to explicitly state that the call should be made from a different phone.

Question 4

- (a) Whilst most candidates stated that the knowledge base holds facts, very few also stated that knowledge base also holds the rules base, and so, relatively few candidates were awarded marks for this question.
- (b) At this level, it is reasonable to expect candidates to be able to describe the use of a knowledge base with a high degree of accuracy. Where candidates gave focussed and precise answers, they did well. However, candidates did not always focus as keenly as was required and gave vague answers that did not warrant any marks. For example, whilst many candidates stated that the knowledge base holds facts, very few stated that these were sourced from experts. Similarly, candidates were aware that rules are used to interrogate the knowledge base, but very few stated that only the relevant rules were used.

Question 5

- (a) Candidates were expected to explain the reasons why calibration is required before sensors are used. It should be noted that the question did not state 'first used', but answers based on calibration for first use were accepted.

The process of explaining a reason is one where candidates identify the reason and then clarify the importance or cause of that reason. Typically, candidates identified a reason why calibration was required but failed to explain this reason. For example, candidates could state that sensors become inaccurate over time. This was a reason for calibration, but the explanation of that inaccuracy is that it occurs due to wear and tear.

- (b) Candidates tended to focus on the differing number of measurements required to complete the calibration. This is a reasonably simple concept, but some candidates confused the issue by discussing criteria, rather than measurements.

When candidates are asked to compare, they are able to identify similarities as well as differences. Few, if any, candidates identified similarities between the two methods.

Question 6

The structure of the question appeared to cause some issues for candidates, some of whom only focussed on one of the two aspects, rather than both. The definition of utility software is quite precise, and so, where candidates attempted to answer this part of the question, they often struggled.

Question 7

Both parts of this question caused some issues for candidates. For **part (a)**, some candidates were aware that the term dynamic meant that things changed, but could not link the concept of change to queries, whilst very few were able to suggest benefits of the use of dynamic queries. Where marks were awarded for **part (b)**, this was typically because a candidate identified that one advantage was reduced file size.

Overall, answers suggested that candidates had little exposure to the use of dynamic queries prior to the examination.

Question 8

For each part, candidates had to describe the process and give a reason for its use for full marks to be awarded. Neither mark available was dependant on the other. Candidates answered with mixed success.

- (a) This part was the most successful of the set.
- (b) This part proved to be a difficult question for candidates, with very few achieving any marks.
- (c) This part was fairly well answered, as most candidates were able to describe it as a method of adding echo to a sound. However, few candidates gave a clear reason why it was used.

Question 9

This question had two demands. Firstly, candidates had to understand the focus of the question, and, secondly, they then had to justify the decision made by the programmer. For the second demand, few candidates attempted to fully justify, with some candidates giving reasons against the decision, and others giving descriptions.

As for the decision to write code that has to be interpreted, candidates showed some awareness that this was a question about the use of interpreters, but as they did not address the need to justify the decision, few gave clear high-scoring answers.

Question 10

Most candidates showed some awareness of the use of real time processing but could not adequately link it to the context of a space rocket. Other candidates simply repeated the wording of the question. Although this suggested a partial understanding of the concept, their answers did not go beyond restating the term '*real-time processing*'. As a result, they could not demonstrate any actual knowledge and therefore gained no marks.

Choosing an appropriate context for a question usually requires considerable forethought. Real-time processing is critically important in the operation of a space rocket, and this is a context with which candidates should be familiar and well-practised.

Question 11

As candidates were asked to describe, rather than state, answers needed to be more than one-word answers. For example, instead of focussing on the reliability, availability and serviceability (RAS) acronym, candidates had to describe what was meant by, for example, reliability.

Very few candidates described characteristics, with some candidates merely identifying, or others answering different questions to that being asked.

Question 12

Many candidates did not realise that this was a different question to the previous one. Therefore, many based their answers on supercomputers. Where candidates scored one mark, this was generally because they knew what the acronym meant.

Question 13

This question was an iteration of a theme that has appeared many times before in this paper. Despite this, marks were still mixed. One common issue was that candidates did not base answers on the context of the question, whilst another was that candidates often gave vague answers. For example, candidates stated that the use of a model was cheaper. There is no basis for this assumption in the information given by the question. However, candidates who stated that the cost of, for example, car repair was avoided, were awarded.

One aspect of modelling that is often used by candidates as an answer, but with insufficient detail, is that of testing with different scenarios. This answer is too vague, as different scenarios can be tested in real life also. The key point about testing with a model is that different scenarios can be created on demand, and do not have to be waited for. As stated, candidates often attempt to give this answer but do not state that the frequency of the scenarios can be manipulated by the model.

Question 14

This question proved to be very little challenge to most candidates.

Question 15

Very few candidates gave any indication that they knew what was meant by the term referential integrity, and even fewer were able to give clear answers about its importance.

INFORMATION TECHNOLOGY

Paper 9626/12 Theory

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Comments on specific questions

Question 1

Most candidates scored at least two marks for this question, with many achieving full marks. Where candidates did not achieve full marks, this was often because they listed issues to do with the train service itself. Whilst answers such as 'the information about the trains may be old' were accepted, answers such as 'the train may not always run on that day' were not.

Question 2

For this question, candidates' answers fell into two broad categories. In the first instance, candidates gave answers that related to checking that data had been correctly transferred. This included that the total is recalculated based on the data that has been received, and the totals are compared to arrive at a conclusion

about the transfer of data. However, in the second, candidates gave generic answers about verification, with very little awareness of the specific topic. Whilst answers of this second type did achieve a mark for showing awareness of checking data, other marks were not available.

Question 3

- (a) As the question asked candidates to give a general description, candidates could make any number of points, with no restriction on the number of points that could be considered as answers. Consequently, where candidates made many points, these answers tended to include a fair number of relevant answers and so candidates scored quite highly for this question. Where candidates correctly identified that the inference engine was the logic centre for the expert system, candidates scored well.
- (b) This question was not as successfully answered as **part (a)**. The mark scheme gave a mark for 'data is entered' and, for many candidates, this was the only mark achieved. Candidates claimed that the knowledge base was created by experts, missing that the knowledge base actually contains the knowledge of experts. A possible mark was available for stating that the knowledge gathered from experts was assessed for relevance, however, very few candidates achieved this mark. Typically, where marks were awarded, this was for candidates stating that data was gathered from experts and entered into the knowledge base using the knowledge base editor.

Question 4

The use of a question to assess the suitability of spreadsheets to use as part of computer modelling was intended to allow candidates to explore their awareness of the limitation of spreadsheets for this specific role. Clearly, spreadsheets may have other roles, but these were not the focus here.

Generally, answers to this question lacked focus and accuracy. Candidates gave generic answers about spreadsheets, typically focussed on cost or describing aspects of spreadsheets that were debatably negative, irrespective of whether they were potential answers to the question. Cost of purchase is rarely awarded as a factor in any question. Similarly, candidates suggested that a negative aspect of a choice is that it may not work. Again, this is rarely accepted as it is a generic answer that could be applied to any scenario.

Where candidates did achieve marks, this was usually because they had identified that the use of spreadsheets for modelling usually required an expert or training. In this context, cost gained a follow-on mark, but only because it is contextually correct.

Question 5

This question asked candidates to identify issues in a piece of pseudocode and to describe what the code as written would achieve. Crucially, candidates were not asked to improve the code.

This question proved extremely difficult to most candidates. Whilst most were able to identify two or three issues with the code, very few were able to describe what the code would achieve, especially as 'it would not work' was specifically excluded by the question.

Where candidates are asked to identify issues with pseudocode, centres are advised that candidates should answer the question in the space provided in the question paper, rather than writing answers on the pseudocode included in the question paper.

Question 6

- (a) This question had many possible answers, but repetition of the theft of data, which was included in the question as an example, was not accepted. Many candidates gave two good answers, but some repeated the 'theft of data' answer and so did not achieve marks. Some candidates gave 'industrial espionage' as an answer. Whilst this is an example of the theft of data, the use of a totally different term, rather than simply repeating that data was stolen, was sufficient for a mark to be awarded.
- (b) A small number of candidates misread the question, and gave advantages. However, many candidates showed some good awareness of the issues.

A small number of candidates attempted to give cost as a negative factor here. As discussed above, this is a generic answer and is also not necessarily correct.

Question 7

- (a) Questions describing the process of creating or planning are common in this paper and are often extremely well answered, especially where candidates can reflect on practical experience to inform their answers. Candidates tended to approach the answer in two different ways. In the first, candidates listed (sometimes by creating a bulleted list) steps that would be taken, whilst others described the steps that had been taken. Either method was acceptable, with the key issue being that candidates had to identify decisions that had to be taken, such as 'choose a room'. For such answers, any indication of choice was acceptable, including decide upon, or identify, as both indicate the narrowing down process from many to one.
- (b) The second section of the question was not as well answered. Most candidates appreciated that the question required reasons in support, although some gave arguments for and against, but some of the reasons given in support were either vague or incorrect.

Question 8

For **parts (a) and (b)** questions, a small number of candidates stated that each was a method of reducing file size. The fact that the same answer could be given for each question suggests that more was required from candidates. No marks were given for this answer.

- (c) Whilst most candidates achieved high marks for this question, some candidates either only focussed on one of the two methods, or did not discuss the impact of each method on the quality of a video file.

Question 9

As included in the introduction to this report, context is extremely important. When answering questions such as this, candidates should consider why the context used has been chosen.

The mark scheme was structured so that full marks could only be achieved where the concept and the context were combined and so where candidates combined the concepts of encryption and pen drives, they generally scored very well. However, the issue of encrypted data being understood versus being read remains. Encrypted data can be read. Encrypted data cannot be understood without first being unencrypted. This is the fundamental concept to do with encryption, and this misunderstanding remains very common.

In addition to the above, this second most common error when candidates are asked about encryption is to describe methods of encryption, despite this not being asked for. This issue was again present in a small number of answers seen.

Question 10

As the question asked for a description, answers had to be more detailed than single word answers. Unfortunately, many candidates either confused mainframe and supercomputers, gave single word answers or acronyms (such as 'RAS'), or gave answers that were incorrect.

Question 11

In answering the question, candidates should identify what the acronym stands for.

A significant proportion of candidates assumed that this question lead on from the previous and so gave answers based on mainframe computers. Had this question focussed on mainframe computers, it would have stated so and would have been question 10(b).

Where candidates attempted to answer by identifying the source of the acronym, a significant proportion of candidates did not know the correct terms.

Question 12

This question was one of the few purely technical questions on the paper. As is common with questions that focus on the technology on which computers work, responses were quite poor. The concept of managing data implies concepts such as maintenance, organisation and access, all of which gave many possible answers.

Some candidates showed good understanding of the role of an operating system, but this was generally focussed on one area, typically the organisation of data, with very few discussing the role in different areas.

Question 13

There were many possible answers that could be given for this question and many candidates described the impacts in good detail.

As the question was asking about impacts of the digital divide, there was no requirement to give a cause of the digital divide or define it.

Candidates should be aware of the need for clarity in their answers. For example, there is a clear difference between stating that there will be a difference in the ability to access health services and stating that advanced health services will be unavailable. Where candidates identified that there would be differences, marks were awarded, but where candidates implied that the issue was constant for everyone, marks were not. The question was about effects of a divide, and so the differences in access were key.

Candidates should also expand on points made, as the question asked for effects. The differences in ability to access health services does mean that people in, for example, poorer regions, are impacted negatively when they have a condition that requires treatment. It is also true to say that this has a negative impact on life expectancy in those areas. This final extension was acceptable as it is a direct consequence of the impact of the digital divide on health services in poorer areas.

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Paper 9626/13 Theory

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Question 15

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INFORMATION TECHNOLOGY

Paper 9626/02 Practical
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Key messages

For this examination, the main issues to note are as follows:

- Candidates must save or export files with the specified file names and file types. File formats and file names not specified in the question paper will **not** be marked.
- Candidates need to make sure the database is normalised to third normal form (3NF).
- Candidates need to make sure that all database reports are formatted as shown in the diagram(s) provided in the question paper.
- Candidates need to be able to manipulate controls within a report including moving controls and their associated labels from one section to another.
- Candidates need to understand the function of a dynamic parameter query.
- Candidates need to understand the structure and components of a pivot table so it can be used to calculate average values.
- Candidates need to be precise in their timings within a video, in terms of splicing clips and adding backgrounds or text to the clips.
- Candidates must follow the instructions on the question paper in relation to **the use of specific** transitions in their video editing.

General comments

Some candidates performed well on much of the paper but did not show in-depth knowledge of normalisation to third normal form (3NF).

Pivot tables proved challenging to many candidates with a significant number of candidates submitting no work for this question.

Most candidates performed well on the video editing task.

Comments on specific questions

Question 1

Most candidates successfully created a database and saved it with the required filename. Many candidates did not create the database normalised as specified in the question paper with a variety of responses seen ranging from single tables to solutions with five or more tables and correct solutions with three tables. Solutions with single tables did not identify that the source file needed to be separated into three tables to contain the customer, robot and model data, so that data was not repeated. Some attempted solutions created a separate table that contained the data for the six sensors. Although there were six different sensor types these should have been held within the robot table. Data that was common to each model such as the device type, actuator type, vision system and microphone should have been placed in the models table to ensure that this data was not repeated.

The field names sometimes contained spaces or candidates were inconsistent in their use of case, some opting to use camel case and some opting to use snake case. Some candidates were inconsistent with a hybrid of the two. Only the Customer table required an ID field as there may be two customers with the same name. Where the three correct tables were created the relationships between the tables were usually created well but when the choice of primary key fields were incorrect the relationship types were not always one-to-many.

The model table was often omitted from candidates' solutions or replaced with a table containing the data for the six sensors. Those that created a model table did not always have the vision system and/or microphone field(s) set as Boolean data and few candidates set appropriate field lengths for the model, device type and actuator type fields. Careful examination of the supplied data file at the start of this question should have indicated that Boolean was the most appropriate data type and the field lengths should have been determined as 7, 6 and 9 characters long respectively. Most candidates who created this table had the 10 correct records with no duplicate values.

Most candidates created a customer table and used appropriate field names although spaces were occasionally seen in the address field names. The telephone field was often erroneously set as a numeric data type rather than a text data type (as calculations would never be performed on a telephone number). Those candidates who correctly set this as a text field rarely set the field length to 12 characters which should have been identified when examining the supplied data file. Most candidates had only the ten correct records with no duplicate data.

The robot table was often created with the serial number set as the primary key. Many candidates identified that the arm length and the six sensor fields should be numeric data types, but far fewer set these as integer sub-types. Few candidates included an appropriate validation routine despite a clear instruction to do so in the question paper.

Question 2

Many candidates successfully created a query and set a search criterion to locate only orders for Ling Enterprises. Far fewer candidates included a correct calculated field for the number of sensors which added the contents of all 6 sensor fields. Many candidates calculated the total number of sensors in the spreadsheet and imported this as a separate field in one of their tables instead of calculating this in the query as instructed on the question paper. Exporting the report in portable document format was completed well but few candidates demonstrated the skills of fitting it to a single page wide and formatting it as shown on the question paper with all required labels and data fully visible.

Some candidates who did not create a database attempted this in other packages but were given no credit as the skills of manipulating the controls in the various sections of the database report were those being tested. Fields were sometimes placed in the wrong order or omitted completely and labels for these fields were not always displayed in a format suitable for the business context. This was noticeable in the use of snake case or camel case in the report labels. The use of snake case or camel case is appropriate for the field names in the database tables but not suitable for a business report to be presented to a manager or for a customer to read. The vision systems and microphone fields were not always displayed as Yes/No with the default tick boxes set by the database package often seen. Controls within the report were not always resized so that all data and labels were fully visible.

Many candidates had been successful at creating the report format as mentioned above, but far fewer were able to add controls to display the customer's name and address in the report header. Many entered the company name and address in text boxes rather than using controls with a data source and few right aligned the address details as shown on the question paper. Most candidates who completed this report successfully sorted this data as specified.

Question 3

Few candidates created a fully functional dynamic parameter query, some had parameters set to input the data, but these were not linked to the fields in the database to restrict the data selected to only those parameters. Two dynamic fields were required; each should have had appropriate text for the input, so that a user could easily understand what data they were required to enter.

This was not always completed with appropriate text and was frequently limited to 'Name' and 'Model' which gave no useful instruction to the user. To test candidates' ability to move controls between different sections of a report the model field should have been moved from the detail row into the report header. This proved challenging to many candidates with some not attempting to move the controls and others copying instead of moving them.

The final part of this question was to model the report with data from two companies and producing the appropriate reports. Few candidates completed this successfully. Many entered the correct data, but the resulting reports did not always show the correct robot records and all the required data in terms of company name, address and telephone number.

Question 4

This question proved challenging for many candidates. Few created a pivot table with the 10 model names as row labels and the six sensors as the columns. Of those candidates who included the correct row and column labels, almost half used the SUM function (which is the default function suggested by the software) rather than the AVERAGE function. Many candidates who completed this as specified used an appropriate number of decimal places although fewer of them removed the grand totals from each column.

Provided a pivot table (of some sort) was located, conditional formatting was credited if it was applied to the data cells within it. Some candidates incorrectly included the row and column labels in their selection range. Many candidates correctly applied a yellow background to their range of cells. Fewer candidates set the rule to the top 1 for all values.

Question 5

Almost all candidates attempted this question with most setting the aspect ratio of the frames to 16:9 even if they did not export in the required 854×480 resolution. Most candidates trimmed the video clip so that it ended at 20 seconds. Many candidates successfully set the background image at 0 seconds with the correct image taken from the first frame of the shark video. The name of the company Tawara Robotics was usually placed in the top right of the frames, although some typographical errors were seen. A small number of candidates erroneously used the name of another company from the database table, such as Ling Enterprises or entered the text 'Name of company' for this. This was frequently sized correctly so that it was clearly a title but did not overlap either shark. Some candidates used transitions for this text despite a clear instruction not to do so.

Candidates often added the subtitle from 3 seconds onwards but there were often typographical errors in the subtitle text. Many candidates did use a smaller font (or unenhanced text) to ensure it was clearly a subtitle and of less prominence than the title. Again, this was often correctly placed so that it did not overlap either of the sharks. Some candidates used transitions for this text despite a clear instruction not to do so.

Most candidates got the timing of 7 seconds correct and at this time removed all text and started the video clip of the sharks. When this clip ended the transition was often applied between this and the turtle clip, but it was not always a dissolve transition. Most candidates did apply the same transition between the end of the turtle clip and the manta clip. Some candidates did not reduce the speed of the manta clip to 0.5 times its original speed meaning their video was often shorter than the required 20 seconds and this was frequently filled with a black background.

The credits were sometimes not displayed while the manta clip was playing even where the timings and reduction of speed were correct. There was little text entry to be placed in the credits, but a significant number of candidates made typographical errors in this text. Sometimes the timing of the scrolled credits meant that the text entered could not be read before it had exited the frame. Most candidates exported the video with the required filename in mp4 format as instructed but some candidates exported in other formats such as .mov, .avi, and .wmv or saved it in in package specific formats such as .wlmv or .pproj. It is essential that candidates submit the files with the filename and file types specified.

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Key messages

Questions are designed to give candidates the opportunity to demonstrate their wider understanding of the syllabus topics to the Examiners. Candidates are expected to have a depth of knowledge of the subject topics. Questions can be set on any, and all, areas of the A Level topic syllabus. Different command words (2025 – 2027 syllabus, p.66) are used enable candidates to show that they can not only recall, select and communicate their knowledge of IT but can also use that knowledge to analyse and evaluate IT issues. The Assessment Objectives (2025 – 2027 syllabus, p.12) clearly show this. It is important that candidates are able to customise their responses according to the command words in the questions so that they can access the full range of marks available.

Centres must ensure that their candidates study the content of all the topic areas. Centres' attention is drawn to the explanatory statements about Topics on p.13 of the 2025 – 2027 syllabus.

Centres must remind their candidates that the number of marks available for each question, shown at the end of the response lines, is an important indicator of how much detail or how many descriptions should be given in the answers. For example, **Questions 1 (a)** has two marks available for describing an autonomous transport system so at least two points need to be made in the answer for full marks.

General comments

Centres are reminded to advise their candidates to answer the question as set and target 'key words' in the question. For example, **Question 4** was about the 'internet of Things (IoT)' and not about the 'internet'.

Candidates must read the whole of each question carefully and apply their knowledge to the scenario in the question set because the full range of marks is only available to candidates for answers referring to the scenario in the questions. For example, **Question 4** was about the impact of the IoT on individuals so answers should not have been about the impact on organisations or healthcare.

Candidates must be encouraged to attempt all questions. Writing a few sentences, even if little is known about the topic, may gain valuable marks towards the overall total. Candidates should also be encouraged to write full sentences and discouraged from writing bulleted, short statements in their responses. Descriptions or explanations can only be conveyed in full sentences. Analyses, discussions and evaluations should also be in full sentences to properly answer the question.

Comments on specific questions

Question 1

This question was about the emerging technology of autonomous transport systems. The wording in the question was carefully chosen to reflect that in the syllabus. Credit was given for generic or specific descriptions.

- (a) Answers could have been about a specific type of autonomous transport system or generic descriptions of such systems but had to include the underlying concepts e.g. are unmanned, integrates technology, uses AI and GPS. However, some candidates confused autonomous transport systems with the transport of *data* or *information*. Some confused the technology with robots and made no mention of transport.

- (b) Credit was given for any valid reference to the use of autonomous transport systems in healthcare, ranging from the transport of patients to the movement of materials such as medicine or hospital wastes. However, examples had to be sensible and feasible. There were some confused answers about robotic surgical techniques which did not gain credit.

Question 2

Grids and guidelines are layout tools used when creating and editing digital images in software. The questions asked for their use so descriptions of what a grid or guideline is or how they are accessed from menus were not required.

- (a) Most candidates answered this question correctly, referring to accurate placement, scaling objects or snapping to grids.
- (b) Most candidates answered this question correctly, referring to accurate alignment and snapping to guidelines.

Question 3

The command word 'compare' indicates that answers must include both similarities and differences. To ensure that candidates understand this, this instruction was added to this question. The additional instruction may not always be included so centres are reminded to ensure that their candidates understand what the command words require in the answers. Answers that did not include both were unable to have access to the full range of marks.

Good answers would have referred to similarities such as both being graphical or visual representations of a project, both showing the routes through a project or showing the critical path along with differences such as CPM being a network diagram and Gantt charts being based on bar charts or CPM not showing a timescale whereas a Gantt chart does.

Question 4

The internet of Things (IoT) is the term given to the interconnection of e.g. smart devices which exchange data using, often, high-speed communication systems. This question asked candidates to consider the impact that this emerging technology is having on individuals. Answers could have referred to e.g. smart homes changing the lifestyles of individuals, wearable computing devices monitoring health and fitness and reporting back to health professionals or the use of smart glasses and augmented reality to assist individuals to navigate around cities. All valid impacts were given credit. There was some confusion over the term with a significant number of answers referring to the impact of the 'internet' and describing the effects of its overuse on health, social media concerns and the hacking of devices. These references did not answer the question and are, possibly, an example of keyword 'spotting' i.e. internet and of not reading the question fully.

Question 5

Weaker answers confused evaluation with testing and described tests that would be performed during development. These gained little, if any, credit. Good answers referred to determining if the database is working as intended, comparing with design, user, system or other specifications and to determine any limitations of the new database.

Question 6

Weaker answers confused 'user requirements' with 'user guide' and described the contents as if they were to explain how to use the system or the authentication of users as they tried to access the database. Answers could have referred to setting out what a user wants from the new system, user requirements being a basis for the contract between user/client and developers or to inform other specifications needed for the development of a new system.

Question 7

Credit was given for correctly defining a virtual currency as e.g. representing an amount of money that is stored only in digital form e.g. coupons or reward points. Many candidates could not precisely define a virtual currency and muddled their answers with vague details of different forms of digital currency e.g. credit cards. Good answers referred to the use of cryptography, no regulation and the lack of oversight along with the

restriction on its use e.g. only within a specific game or retail group. As clearly stated in the question, advantages and disadvantages were required so answers that did not include both could not be awarded full marks.

Question 8

- (a) Weaker answers referred to preventing the spread of viruses or malware without a mention of assessing the risk. Good answers would have referred to a determination of the extent of a threat and the use of this assessment to attempt to mitigate it.
- (b) Quantifying a risk means measuring or judging the amount of the risk. The amount is often represented as a number. While a few answers noted that a value could be placed on the possible financial damage that could be caused if a disaster happened, few answers referred to assigning a numerical value or scale to the risk.

Question 9

Networking is a large part of this syllabus so it was expected that this question would be answered well. However, there were very few good answers. Many answers lacked precision or were factually wrong. Centres should ensure that their candidates study all the topics in the syllabus. This type of question requires precise and technically accurate details to be given.

- (a) Many factually incorrect answers were seen with descriptions of IP addresses being assigned by routers or being assigned at the manufacturing stage. Correct references to the NIC 'looking for' a DHCP server, the use of UDP for communications or the exchange between the NIC and the DHCP server were rarely seen in the responses.
- (b) Many vague and factually incorrect answers were seen. A good answer would have referred to the uniqueness of the IP address being used to identify a NIC/device, the network to which it is connected and how this allows data packets to be exchanged.

Question 10

- (a) The question required candidates to describe 'how' the JavaScript code works, meaning descriptions of the code without describing how it operates did not gain credit.
- (b) A description of an HTML *event* was required, meaning JavaScript functions or methods did not gain credit.

Question 11

Most candidates were able to describe the use of email by governments; however, responses that consisted only of lists or basic descriptions without further explanation, commentary, or judgment did not achieve the higher mark range. The command word 'evaluate' requires candidates to assess the importance of the use of email by governments (see p.66 of the syllabus). For example, while governments use email to distribute information to citizens, a strong response would go further by making judgments, such as, emails can be targeted to specific groups, ensuring that only those who need the information receive it, which makes communication more effective. High-quality answers should move beyond simple descriptions and include evaluative comments that demonstrate critical thinking about the facts presented.

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Key messages

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Centres must ensure that their candidates study the content of all the topic areas. Centres' attention is drawn to the explanatory statements about Topics on p.13 of the 2025 – 2027 syllabus.

Centres must remind their candidates that the number of marks available for each question, shown at the end of the response lines, is an important indicator of how much detail or how many descriptions should be given in the answers. For example, **Question 1** has four marks available for describing the structure of a cellular network so at least four points need to be made in the answer for full marks.

General comments

Centres are reminded to advise their candidates to answer the question as set and target 'key words' in the question. For example, **Question 3** was about the 'internet of Things (IoT) and not about the 'internet'.

Candidates must read the whole of each question carefully and apply their knowledge to the scenario in the question set because the full range of marks is only available to candidates for answers referring to the scenario in the questions. For example, **Question 3** was about the impact of the IoT on the environment so answers should not have been about the impact on individuals.

Candidates must be encouraged to attempt all questions. Writing a few sentences, even if little is known about the topic, may gain valuable marks towards the overall total. Candidates should also be encouraged to write full sentences and discouraged from writing bulleted, short statements in their responses. Descriptions or explanations can only be conveyed in full sentences. Analyses, discussions and evaluations should also be in full sentences to properly answer the question.

Comments on specific questions

Question 1

Questions that ask for technical details require detailed answers. Some good descriptions of the basic structure of cellular networks were seen but most answers lacked precision. Credit was given for answers that correctly described base stations using low power radio transmissions over a range of frequencies, the use different frequencies for adjacent stations and the varying number of stations in different geographical scenarios.

Question 2

- (a) Many candidates included the order of tasks but there were few other valid points in most answers. A workflow shows e.g. the steps needed to complete a task, the resources needed and the team members needed.
- (b) This question was not well answered. Most answers did not address how a workflow would be used. A good answer could have included e.g. to show the critical path, where the resources should be allocated or to work out the cost of each task.

Question 3

While there some good answers, many candidates confused the internet of Things (IoT) with the 'internet' and wrote about how use of the internet affected people. Candidates must read the question carefully. Good answers related the increase in inter-connected devices to the increased use of electrical power generated by burning fossil fuels, increased e-waste or more precise control over e.g. home appliances.

Question 4

- (a) A precise answer describing a datafile containing related data was required for the mark.
- (b) Some good answers to this question were seen. As well as being able to use pivot tables, candidates should also understand why they are used. However, many answers lacked detail and were vague. Answers should have included concise details of e.g. the ability to generate multiple reports from the same dataset and the rapid discovery of patterns in complex data.

Question 5

Networking and communication are a large part of this syllabus so it was expected that candidates would be able to answer these questions well. However, many answers lacked detail or were factually wrong. Candidates are expected to have a knowledge of all the topics in the syllabus.

- (a) Most candidates missed the point about message switching in that there were few references to the whole message being transmitted at once or to the 'store and forward' method of transmission.
- (b) The reasons why message switching is used were not well explained by most candidates. Answers should have referred to e.g. the efficient use of network bandwidth, that messages can be broadcast to multiple destinations simultaneously to avoid repeating the message or that the message is stored on intermediate nodes so attempts at sending can be repeated.

Question 6

This question required candidates to include both the advantages and the disadvantages of using Wi-Fi instead of copper cables for then network connections of sales terminals in a supermarket. Answers that did not include both could not be awarded the full marks. Good answers expanded on factual points and would have included e.g. the ability to place sales terminals anywhere without the physical restrictions of running cables or the susceptibility to wireless transmissions from other devices.

Question 7

Answers that restated the JavaScript code without explaining how it produces the list did not gain credit. Credit was given for step-by-step explanations of how the code works to create a list e.g. explaining the use of the variables, explaining the use of the line break (
) and explaining how the contents of the variable *paper* is passed back to the HTML.

Question 8

Candidates were expected to describe in some detail how to control the speed of the movement of objects in computer animations. Most candidates answered by inaccurately describing the use of key frames. Good answers described the use of timelines, the altering the number of frames or the spacing of the objects within a frame.

Question 9

Many answers were not given credit because they were little more than repeats of the information given in the question. Good answers pointed out that e.g. a central bank digital base money is guaranteed by central banks and the currency can be considered as legal tender, has its value set against fiat currency and can be transferred from peer to peer.

Question 10

Most candidates could describe the use of BitTorrent, but answers were often vague and inaccurate.

- (a) A few good answers with descriptions of the process were seen but most lacked detail.
- (b)(i) Most candidates answered this question with correct references to malware, use of network bandwidth and the possible infringement of copyright laws due to the download of 'pirated' materials or games.
- (ii) Most candidates answered this with a reference to the use of a VPN, but some inaccurately suggested that encryption alone would suffice.

Question 11

As with similar questions requiring both benefits and drawbacks, both were required for access to the full range of marks. While most answers lacked sufficient detail about most of the points made, some candidates were able to add some expansion and additions to their basic points. In questions that require candidates to 'discuss', points should be made and then expanded with further detailed comments (see p.66 of the syllabus for the command word definitions). Also, few candidates could accurately describe direct changeover in full as being when the old system is stopped and the new system is started and used immediately.

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Key messages

It is essential that centres advise their candidates to pay particular attention to matching all the specifications provided in assessment tasks. Achieving full credit requires candidates to follow the instructions closely, ensuring that every detail is addressed correctly. Subtle differences in specification can significantly influence the accuracy and appropriateness of responses. By thoroughly identifying and addressing these finer points, candidates will be able to maximise their performance across all sections of the examination.

General comments

Many candidates could not demonstrate their full potential due to their difficulty with the problem-solving and interpretation elements of the tasks.

Comments on specific questions

Task 1 – Data analysis and visualisation

This task required candidates to prepare, analyse, aggregate, and visualise wind direction data from the WeatherData.csv file to help fruit farmers plan windbreaks. It was a very straightforward task involving only cleaning the dataset, classifying wind directions, counting occurrences, and presenting the results in a comparative chart that highlighted how often each wind direction occurred (both gusts and main winds).

Cleaning the dataset only required candidates to hide unneeded columns. The instruction to hide the columns was explicit in the question paper but many candidates deleted those columns. Fortunately, there were no follow up tasks that needed the deleted data but following instructions precisely is crucial at this level.

The next step was to use the provided angle ranges to categorise the wind direction data into compass directions (north-easterly, south-easterly, etc.) and create a table that counted the number of days for each of the four wind directions, distinguishing between the counts for 'highest wind gust' and 'main wind direction'.

This involved COUNTIF() formulae with inequalities set as shown in the angle ranges provided. For example, the number of occurrences for the North-east direction listed in column 'C' could be determined by use of the formula '=COUNTIF(C:C,">=90")-COUNTIF(C:C,">179")'. Other valid solutions were acceptable provided the ranges used matched those given.

Few candidates were completely successful in this part of the task. The skill and experience needed was minimal and not beyond IGCSE level, so it must be assumed that the difficulty lay with the problem-solving element and interpretation of the task. At 'A2' level, the data handling skills required are no greater than those tested at 'AS' level. The discriminating factor is the problem-solving ability of the candidates. Centres must recognise the need to prepare candidates with this element as a priority.

The visualisation component of the task required candidates to produce a chart that could provide easily interpreted information for a lay user. If candidates managed complete the table, whether accurately or not, a single keystroke could have generated the basic chart and the addition of an explanatory title would have completed the task.

Task 2 – Vector graphics

Candidates were required to produce close copies of the images of apples shown in the question paper. The body of the apple needed to occupy a 300-pixel square and the colour was specified as RGB 180,0,0.

The first part was done well. A few candidates did not size the shape correctly, but since the shape usually had the correct proportions, it may be that candidates did not believe the size specified was as important as the shape.

Candidates needed to examine the images shown in the question paper very closely. For example, in the whole apple image, the dimple where the stalk is connected to the apple skin needed to be convincing and the best way to achieve this was to use a gradient. An outlined shape was not satisfactory. Similarly, the leaves shown had curves that need to be matched by having a simple curve on one side and a compound curve for the other.

Most candidates produced fair attempts for the bites, but one common error was that candidates created a shape of the bite and used it to overlay the whole apple. This became obvious when the image was placed on a coloured background. A proper solution to these parts of the task was to node edit the apple or to use an acceptable shape as a cut out.

Centres should advise candidates that, in order to gain full credit throughout tasks like these, they must match all the specifications shown or described very closely and identify subtle details that could affect their answers.

Task 3 – Animation

Candidates animated the apple being 'bitten' well, by replacing the images every two seconds. The only marks that candidates were likely to lose were if the images were not placed or sized as shown and if the images did not appear to remain static as they were replaced.

Step 5, however, required candidates to 'tween' the stalk and leaves turning brown and falling to the bottom of the frame. Solutions that were awarded more credit showed the leaves taking two seconds to change to brown and falling following the path shown over another second. Fully satisfactory solutions were not often seen, and many candidates seemed to find the change in colour difficult since many attempts showed the leaves changing from green to brown instantly. It seems that while candidates were familiar with simple tweens, they were less comfortable with the manipulation of colours.

In general, it was clear that nearly all candidates were competent with animation but again sufficient adherence to the specifications given was not often seen.

Task 4 – Programming for the web

There were four parts to this task and several possible code alternatives. While full credit could be gained for fully working solutions there were marks to be gained for valid code and efficiency when the code for non-working solutions was examined.

Candidates had to complete the `addData()` function in the script section of the html page provided. The purpose of the function was to populate an array. The function to display the array was provided in the script.

It was clear that although not many candidates were familiar with working with arrays, most were able to appreciate the need to create the variables, and many were able to add the code to test whether all the inputs had been filled.

The main issues noted were:

- All the necessary code should have been added within the `addData()` function. As long as the variables had been declared within the `<script>` tags, the code could be valid, but centres might stress the difference between using Global and Local variables.
- When declaring variables candidates often forgot to declare a variable for the 'day'.
- Many did not add the `value` property accessor to the variables they declared.

- When attempting to clear the inputs, many candidates confused the use of the `.value` property accessor and the `.innerHTML` property of the DOM element which was inappropriate in the code to populate the array and clear the inputs.

In general, it seemed that while most candidates made fair attempts at parts of the task, few had the level of skill and experience to produce complete solutions. In particular, the use of arrays was seen as an area for centres to address.

Key development areas for centres to support candidates

To strengthen learner needs noted in this session's tasks, centres are advised to focus on the following:

- prioritise problem-solving and interpretation skills alongside technical competence
- train candidates to follow specifications exactly and recognise subtle details
- provide targeted practice in arrays, formulae, colour manipulation, and vector editing
- emphasise clarity and usability in outputs (charts, graphics, animations, code).